Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Personalized Context-Aware AI Assistant for Adaptive Communication and Everyday Task Support			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication (√), Advanced Robotics & Manufacturing (), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Human Resource Competency Development Services	Human–System Interaction Technologies	*Design Services*	Design · Multimedia Design Technologies
Concept of Proposed Technology	The proposed technology aims to develop **a user-personalized, context-aware AI assistant** that supports everyday communication and task execution by adapting its responses to individual user preferences, goals, and situational contexts. The system is explicitly designed from a human-centered AI perspective, treating personalization as an interaction design problem rather than a purely model-centric optimization task. Unlike conventional large language model–based chatbots that provide in-context responses but do not explicitly account for user-specific			

	preferences and goals, the proposed system dynamically adjusts its reasoning depth, tone, and information structure through preference- and goal-centered user modeling and adaptive persona mechanisms. The AI assistant is designed to help users better understand, summarize, and communicate information encountered in daily life, including professional meetings, healthcare-related information, and routine planning tasks. By aligning information presentation with the user's intent and cognitive preferences, the system reduces information overload and improves comprehension and decision-making efficiency. **Key Features** ● Preference- and goal-centered user modeling for personalized dialogue generation using LLM ● Adaptive persona-based response generation according to usage context (e.g., work, healthcare information, daily planning), **with responses further tailored using user-specific preferences and goals relevant to each context** ● Context-aware understanding and summarization of complex information ● Human-centered AI support for everyday communication and task completion, with interaction strategies informed by user comprehension and controllability considerations ● **Design and execution of controlled user studies, led by the research institute, to evaluate information understanding, interaction efficiency, and perceived controllability of adaptive AI interactions** ● **Analysis of existing user–chatbot interaction logs and in-depth user interviews to derive design guidelines and actionable insights for preference- and goal-centered user modeling in personalized dialogue generation**
Rationale for Joint R&D Project	The realization of a highly personalized and deployable AI assistant requires close collaboration between advanced AI research institutions and industrial partners. A joint R&D framework is essential to translate an adaptive personalization system into scalable, real-world AI services while addressing system robustness, privacy, and usability constraints. ● The research institute contributes expertise in user modeling, adaptive AI systems, and context-aware dialogue systems. ● The Korean SME partner contributes experience in AI service deployment, user-facing product development, and system scalability. ● Joint validation is required to ensure technical feasibility and commercialization readiness.

	Expected benefits of joint R&D: ● **Accelerated development of deployable personalized AI technologies** through the integration of advanced AI systems and real-world service validation. ● **Reduced technical and interaction risks** via iterative refinement of adaptive persona and personalization mechanisms based on empirical user data. ● **Creation of actionable design guidelines** for preference- and goal-centered user modeling applicable to personalized AI services. ● **Enhanced commercialization readiness of Korean SMEs** by providing validated personalization technologies and design insights for AI-based products.
Key Words	1. Personalized AI Assistant 2. Adaptive Persona Modeling 3. Context-Aware Dialogue and Summarization 4. Human-Centered Everyday Task Support

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The objective of this project is to develop a deployable personalized AI assistant that demonstrably improves user understanding and task efficiency through adaptive communication technologies. Specific objectives include: ● Development of a preference- and goal-based user modeling system ● Development of adaptive persona selection and dialogue generation mechanisms ● Development of context-aware summarization techniques for everyday information ● Validation of personalization effectiveness through quantitative user studies			
R&D Target Specifications	Personalized response alignment accuracy ≥ 85% Context-appropriate persona selection accuracy ≥ 90% User information understanding improvement ≥ 30% compared to non-personalized baselines Reduction in repeated user inputs ≥ 30%			
R&D Plan	**Year 1: Core System Development and Simulation Validation** ● Design of user preference and goal modeling framework ● Development of baseline adaptive persona mechanisms ● Implementation of context-aware dialogue and summarization system ● Simulation-based evaluation of personalization accuracy **Year 2: System Integration and User-Centered Optimization** ● Integration of personalization modules into a unified AI assistant prototype ● Optimization for real-world usage constraints (latency, stability, privacy) ● Design and execution of controlled user studies to evaluate information understanding, interaction efficiency, and perceived controllability of adaptive AI interactions ● Iterative refinement of personalization strategies **Year 3: Real-World Deployment and Performance Validation** ● Deployment of the AI assistant in real usage scenarios ● Large-scale user evaluation to validate quantitative performance targets ● System reliability and scalability assessment ● Preparation for technology transfer and commercialization			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	● Analysis of existing user–chatbot interaction logs and in-depth user interviews to derive design guidelines and actionable insights for preference- and goal-centered user modeling in personalized dialogue generation ● Development of core personalization, adaptive persona, and a context-aware dialogue system ● Design and execution of controlled user studies to validate personalization effectiveness and interaction outcomes
Expected R&R of Korean SME Partner	● Integration of AI technologies into user-facing service platforms ● Data collection and system testing in real-world environments ● Product-level optimization, deployment, and commercialization planning

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed project is expected to deliver a prototype personalized AI assistant that significantly improves user understanding and task efficiency in everyday communication scenarios, while also generating empirical design insights into how users interpret, adapt to, and collaborate with adaptive AI systems. The project will advance human-centered AI technologies focused on personalization and adaptive interaction. From an industrial perspective, the outcomes will provide Korean SMEs with core personalized AI technologies applicable to productivity tools, communication platforms, and next-generation AI services. The project will support commercialization readiness and strengthen the competitiveness of Korean SMEs in the global AI service market through international collaboration .

6. References

(1) Kaely Hall, Max Ohsawa, Vedant Swain, Jennifer G Kim, “ALL.me: Supporting Identity Work through Human-AI Alignment,” ACM Conference on Human Factors in Computing Systems (CHI) 2026

(2) Ryuhaerang Choi, Taehan Kim, Subin Park, Jennifer G Kim, Sung-ju Lee “Private Yet Social: How LLM Chatbots Support and Challenge Eating Disorder Recovery. ,” ACM Conference on Human Factors in Computing Systems (CHI) 2025

(3) Kaely Hall, Marcus Ma, Xinyue Zhang, Vedant Das Swain, Jennifer G Kim “Understanding Human-AI Misalignment in LLM-Based Job-Seeking Support for Neurodivergent Users,” International ACM SIGACCESS Conference on Computers and Accessibility 2025

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization